

The background features a light blue gradient with abstract circuit-like patterns. Purple and orange lines, some straight and some curved, crisscross the frame. Small circles, some solid and some hollow, are placed at various points along these lines, resembling nodes or components in a network. The overall aesthetic is clean and modern, typical of a technical presentation.

DevOps, tests et déploiement

Introduction aux concepts fondamentaux



Programme

01

DevOps, Docker & Sécurité

- Introduction
- Rappels Git
- Docker & Compose
- Sécurité

02

Tests, CI & analyse statique

- Tests automatisés
- Analyse statique de code
- Intégration continue
- Dépendances et sécurité

03

Déploiement, CI, CD & monitoring

- Déploiement
 - CI/CD
 - Gestion des logs / monitoring
- 



01

DevOps, Docker & Sécurité



Qu'est-ce que DevOps ?

Development & Operations

Qu'est-ce que DevOps ?

Development & Operations

Qu'est-ce que DevOps ?

Development & Operations

Orientation client

Livraison continue de
version mineures

Collaboration

Entre les équipes
opérationnelles et de
développement

Automatisation

Rapidité et solidité des
processus répétitifs

Responsabilité

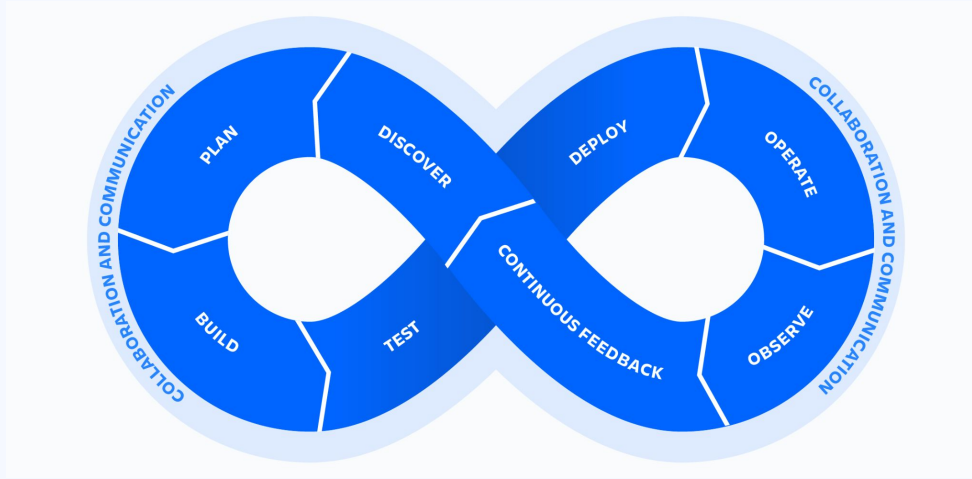
Implication globale et
partage des
responsabilités

Rapidité

Livraisons plus rapides
et régulières

Qu'est-ce que DevOps ?

Development & Operations



Qu'est-ce que DevOps ?

Development & Operations

Culture globale d'entreprise visant à faire **collaborer** très étroitement les équipes de développement et équipes opérationnelles

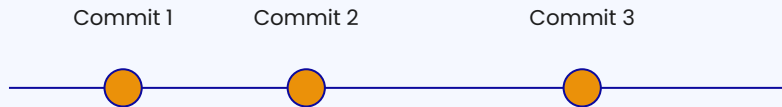
Ensemble d'outils permettant d'efficacement **déployer** et **surveiller** des applicatifs.

Ensemble de processus permettant d'améliorer le **développement**, la **livraison**, la **surveillance** et la **prise de retours** de produits

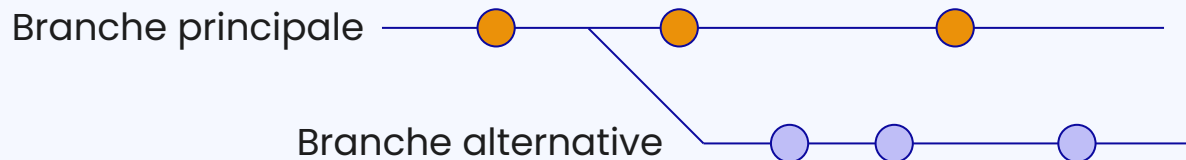
Rappels Git



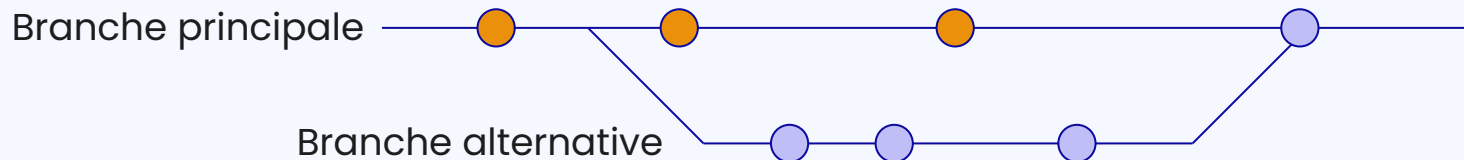
Système de contrôle de versions



Système de contrôle de versions



Système de contrôle de versions

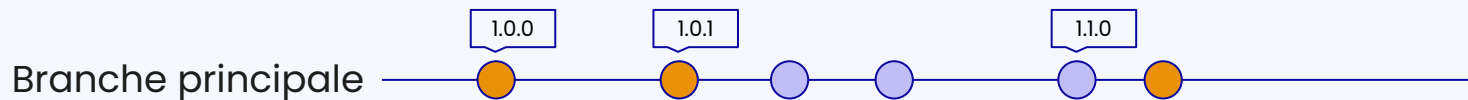


Système de contrôle de versions

Branche principale —●—●—●—●—●—●—



Système de contrôle de versions



Rappels Git

Conseils

- Utilisez une convention de nommage pour les commits

Semantic-commit : `feat(login): added login feature`

Liez vos commits à l'ID de tâches

- Faites des commits atomiques

Un travail = un commit

- Utilisez une convention de nommage pour les versions

Semantic-versioning : `1.0.0 - 1.0.1 - 1.0.1-rc.0`

Suivez à la lettre l'incrément de versions

Ne créez pas de versions, incrémentez la

Rappels Git

Conseils

- Utilisez une convention de nommage pour les branches

Exemple : feat/ID-120, release/1.0.1

Liez vos branches à l'ID de tâches

- Gardez une historique clair

Choisissez une méthode de merge de MRs adaptée

Définissez des noms de commits de merge informatifs

- Supprimez les branches inutiles

Docker

Systeme de conteneurisation

Système de conteneurisation

Conteneur

sorte de machine virtuelle ultra
légère, permettant de pouvoir
exécuter des applications

Système de conteneurisation

Conteneur

sorte de machine virtuelle ultra légère, permettant de pouvoir exécuter des applications

Isolation des applicatifs

Portabilité cross-OS

Hautes performances

Scalabilité

Docker

Docker

Docker

Docker Engine

Moteur du système Docker,
permet de créer et exécuter les
conteneurs

Docker Daemon

Traite les requêtes effectuées à
Docker

Docker Client

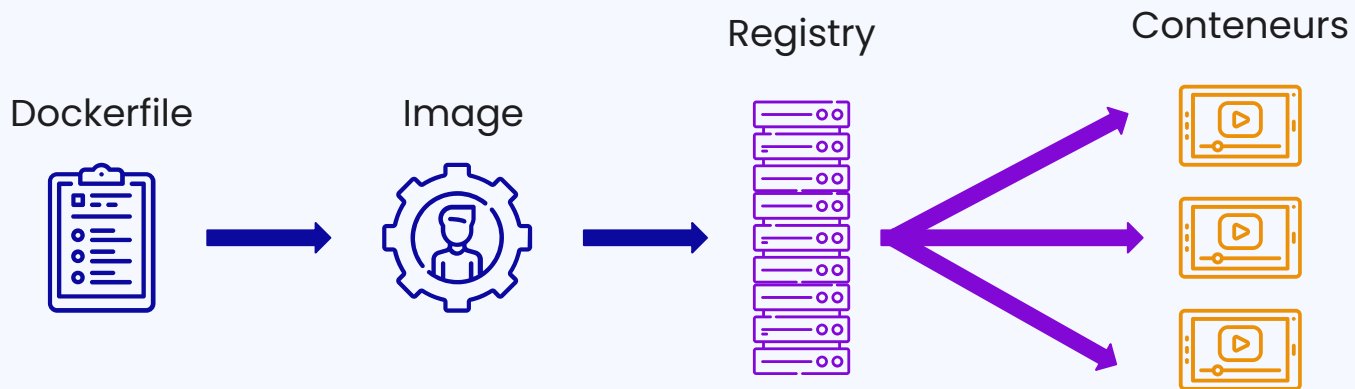
Interface de communication
avec le système Docker

Docker

Docker

Installation

Docker



Docker

Docker

Atelier pratique

Compose

Docker Compose



Programme

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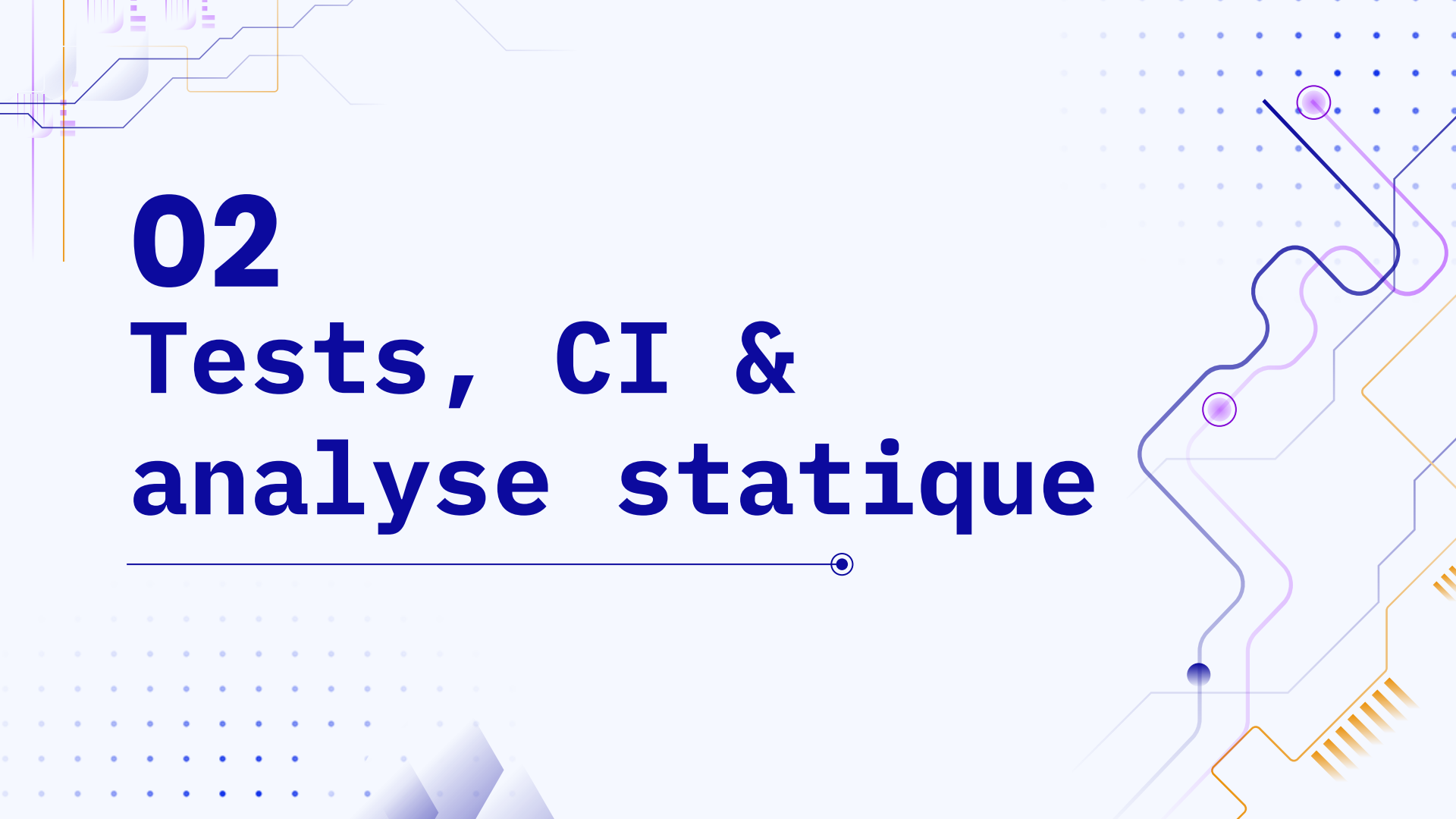
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02

Tests, CI & analyse statique

Qu'est-ce qu'un test automatisé ?



Qu'est-ce qu'un test automatisé ?

Script de test

Qui **vérifie** si le résultat d'actions
Est **conforme** aux attentes

Pourquoi automatiser ?



Pourquoi automatiser ?

- Gain d'**efforts**
- Gain de **temps**
- Réduction des **coûts**
- Amélioration **qualité**
- Répétabilité
- **Couverture** + large
- Tests en **continu** (CI)

Les différents types de test

Test unitaire

Test d'intégration

**Test de bout en bout
(End-To-End - E2E)**

Test unitaire



Test unitaire

Caractéristiques

Teste une petite portion de code (niveau fonction)
Simule les dépendances “externes” (mocks / stubs)

Avantages

Exécution rapide
Pas besoin d’environnement

Inconvénients

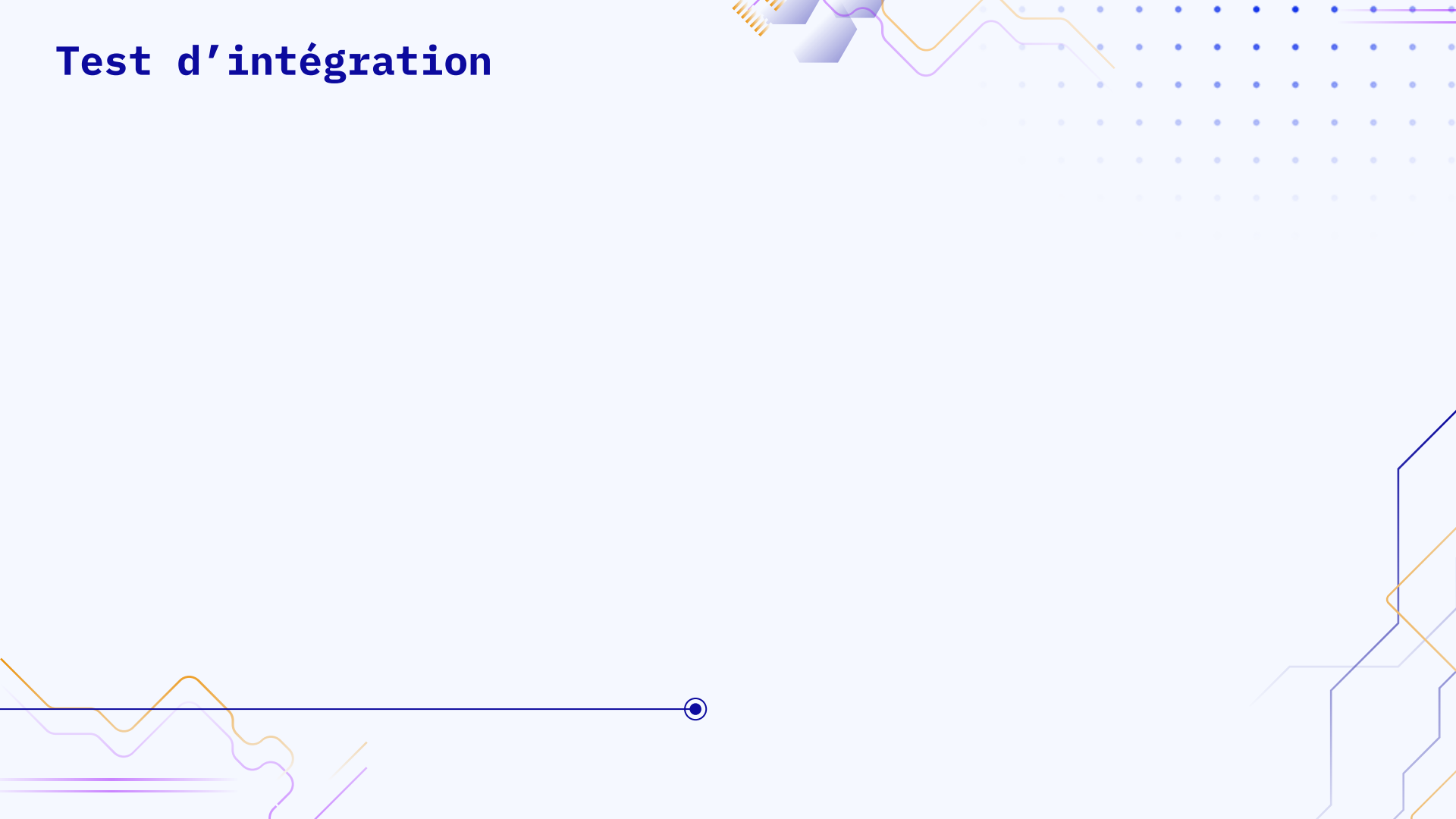
Ne reflète pas l’usage “réel”

Quel usage ?

Large couverture du code et de toutes les
possibilités techniques



Test d'intégration



Test d'intégration

Caractéristiques

Teste le bon fonctionnement de composants de code
Ne repose pas uniquement sur des "mocks"

Avantages

Reflète un usage "réel" des
composants techniques

Inconvénients

Plus long que les TU
Nécessite un environnement

Quel usage ?

Vérification de la bonne intégration des
composants entre eux (services, BDD, APIs..)



Test de bout en bout



Test de bout en bout

Caractéristiques

Teste le bon fonctionnement de l'application en conditions réelles
Repose uniquement sur des composants réels et déployés

Avantages

Reflète l'usage métier de
l'application

Inconvénients

Tests les plus longs
Coûteux à maintenir et évoluer

Quel usage ?

Test orienté métier / fonctionnel pour vérifier le
bon fonctionnement de l'application en conditions
réelles



Test de bout en bout

Caractéristiques

Teste le bon fonctionnement de l'application en conditions réelles
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Inconvénients

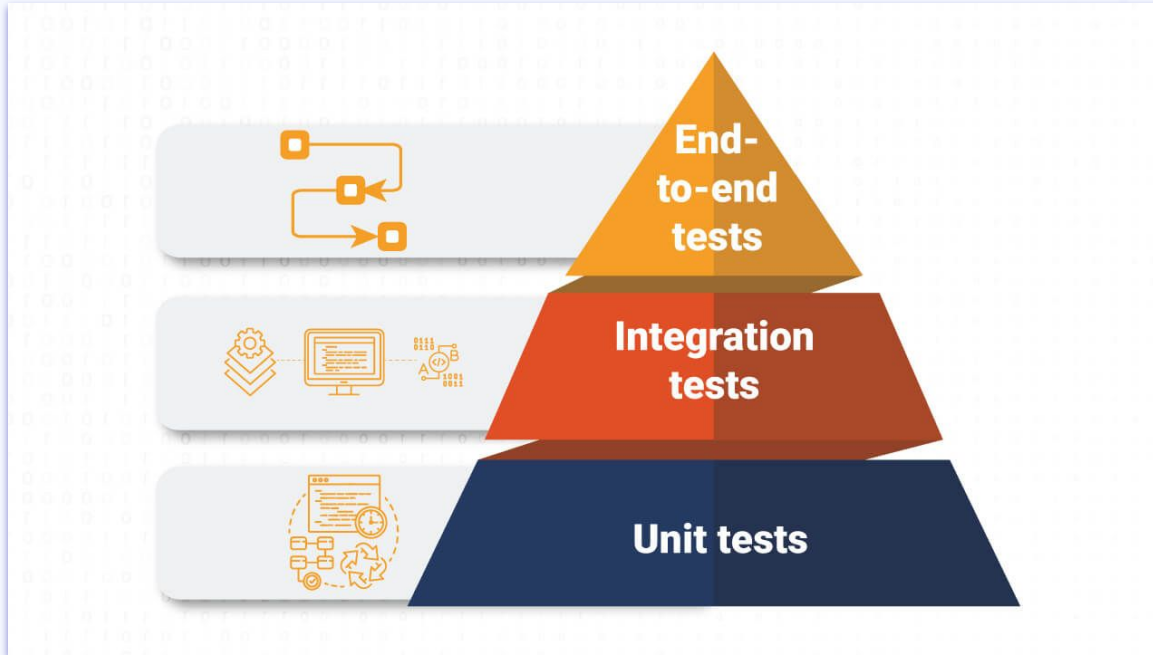
Tests les plus longs
Coûteux à maintenir et évoluer

Quel usage ?

Test orienté métier / fonctionnel pour vérifier le
bon fonctionnement de l'application en conditions
réelles



La pyramide des tests



AAA Rule :

- **Arrange** : préparer le contexte
 - **Act** : exécuter l'action ou la fonction à tester
 - **Assert** : vérifier les résultats obtenus
-
- Écrire des tests courts et précis
 - Tester une seule chose à la fois
 - Faire des tests indépendants



Tests

Tests

Atelier pratique

Analyse statique de code



Pourquoi analyser son code ?

- Amélioration de la qualité du code :
 - *respect des bonnes pratiques et standards*
 - *harmonisation du style de codage*
 - *meilleure maintenabilité sur la durée*
- Détection précoce des erreurs :
 - *bugs potentiels*
 - *problèmes de sécurité*

Analyse statique de code

Analyse statique de code

Atelier pratique



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03

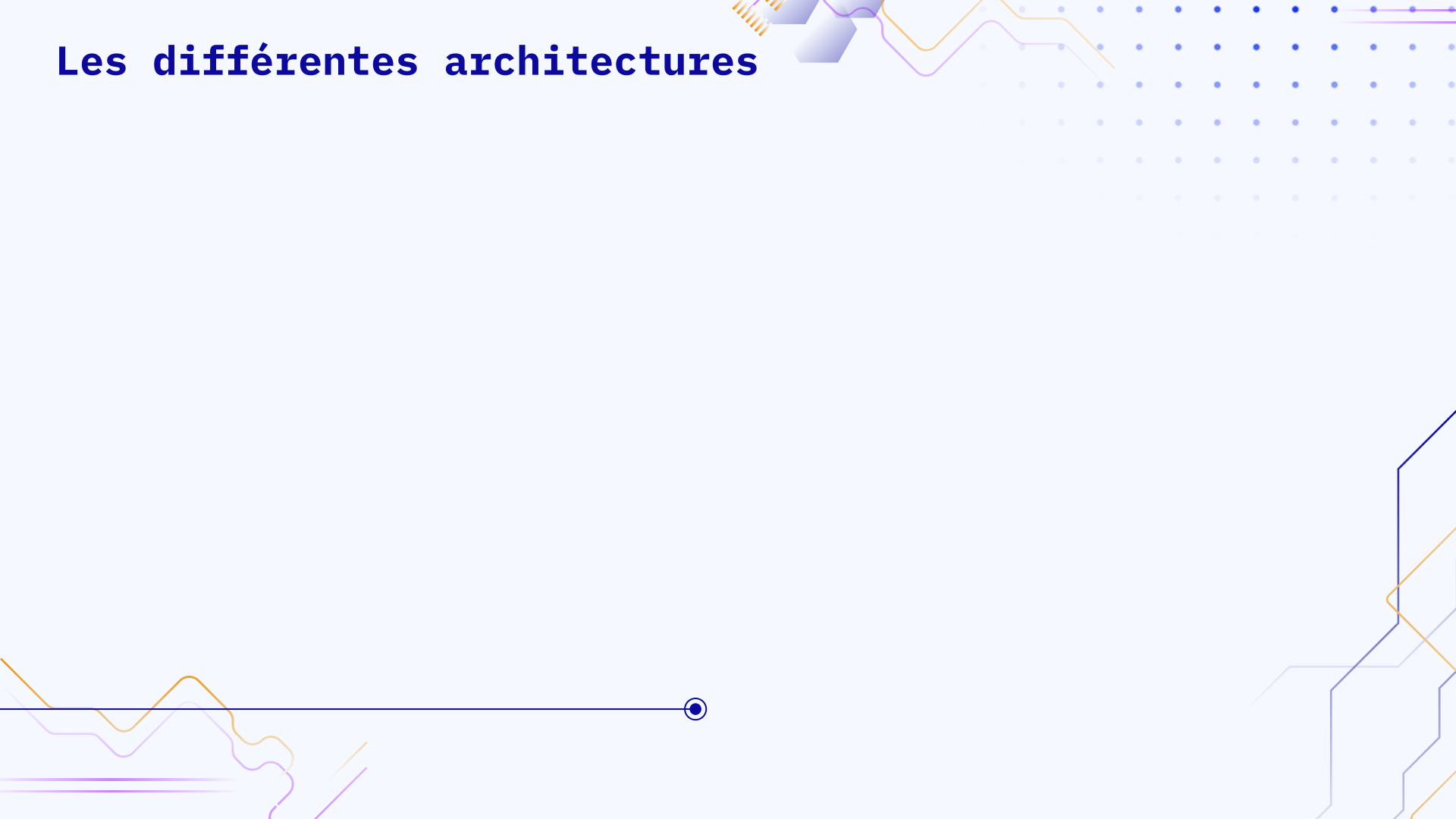
Déploiement, CI, CD & monitoring



Que signifie déployer ?

**Mise à disposition
d'un applicatif
en ligne**

Les différentes architectures



Les différentes architectures

Serveur classique "monolithique"

Une machine (physique ou VM)
Une instance de l'application

Avantages

Simple à mettre en place
Facile à comprendre

Inconvénients

Scalabilité limitée
Risque élevé en cas de panne
Maintenance complexe

Les différentes architectures

Serveur multi-services conteneurisés

Une machine exécutant des services conteneurisés
Peut accueillir plusieurs instances distinctes

Avantages

Séparation des services
Isolation des environnements

Inconvénients

Scalabilité statique
Le serveur reste un point unique de défaillance

Les différentes architectures

Serveurs avec orchestrateur de conteneurs

Une ou plusieurs machines accueillant un orchestrateur
Accueillent et gèrent dynamiquement les instances

Avantages

Scalabilité dynamique
Haute disponibilité
Automatisation-friendly

Inconvénients

Complexe à mettre en place

Et le cloud ?

Que sont AWS, Azure, GCP ?

Infrastructures

**Environnements
d'hébergement**

Peut accueillir n'importe quelle
architecture

Mets à disposition différents
services

Peut faciliter la mise en place
de certains systèmes

Les différentes stratégies



Les différentes stratégies

Stratégie	Downtime	Rollback	Complexité	Pour ?
Direct	Oui	Difficile	Facile	Interne, petits projets, faible trafic

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Canary	Variable	Facile	Élevée	Grande échelle

Les différentes stratégies

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Rolling	Non	Moyen	Moyenne	Clusters & micro-services

Les différentes stratégies

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Comment déployer ?

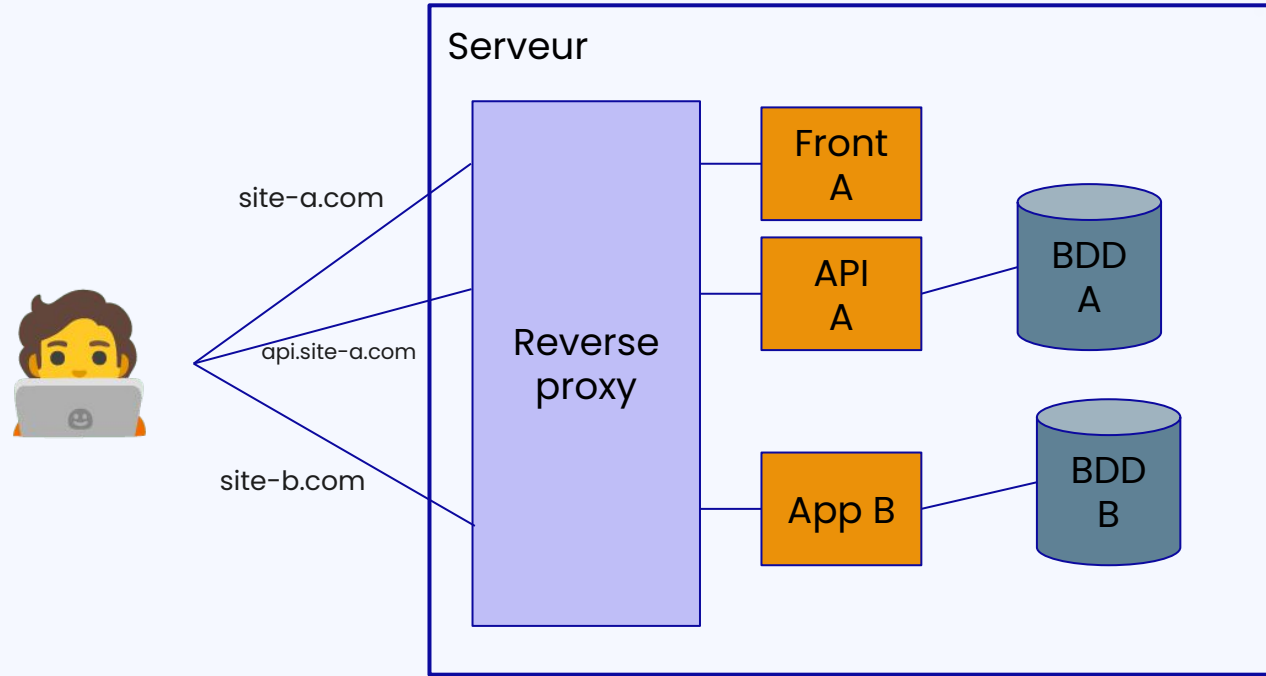


Comment déployer ?

Cas basique

GIT
+ Docker
+ Docker Compose

Serveur avec conteneurs



Continuous **Integration**

Intégration continue

Continuous **Delivery**

Livraison continue

Continuous **Deployment**

Déploiement continu

CI/CD, pourquoi ?

**Automatiser les étapes
de validation
et de mise en production
de nouveau code**

Faciliter l'intégration de nouveau code vers une **branche partagée**

Analyse statique de code

Audit de sécurité

Tests automatisés

Build des applications

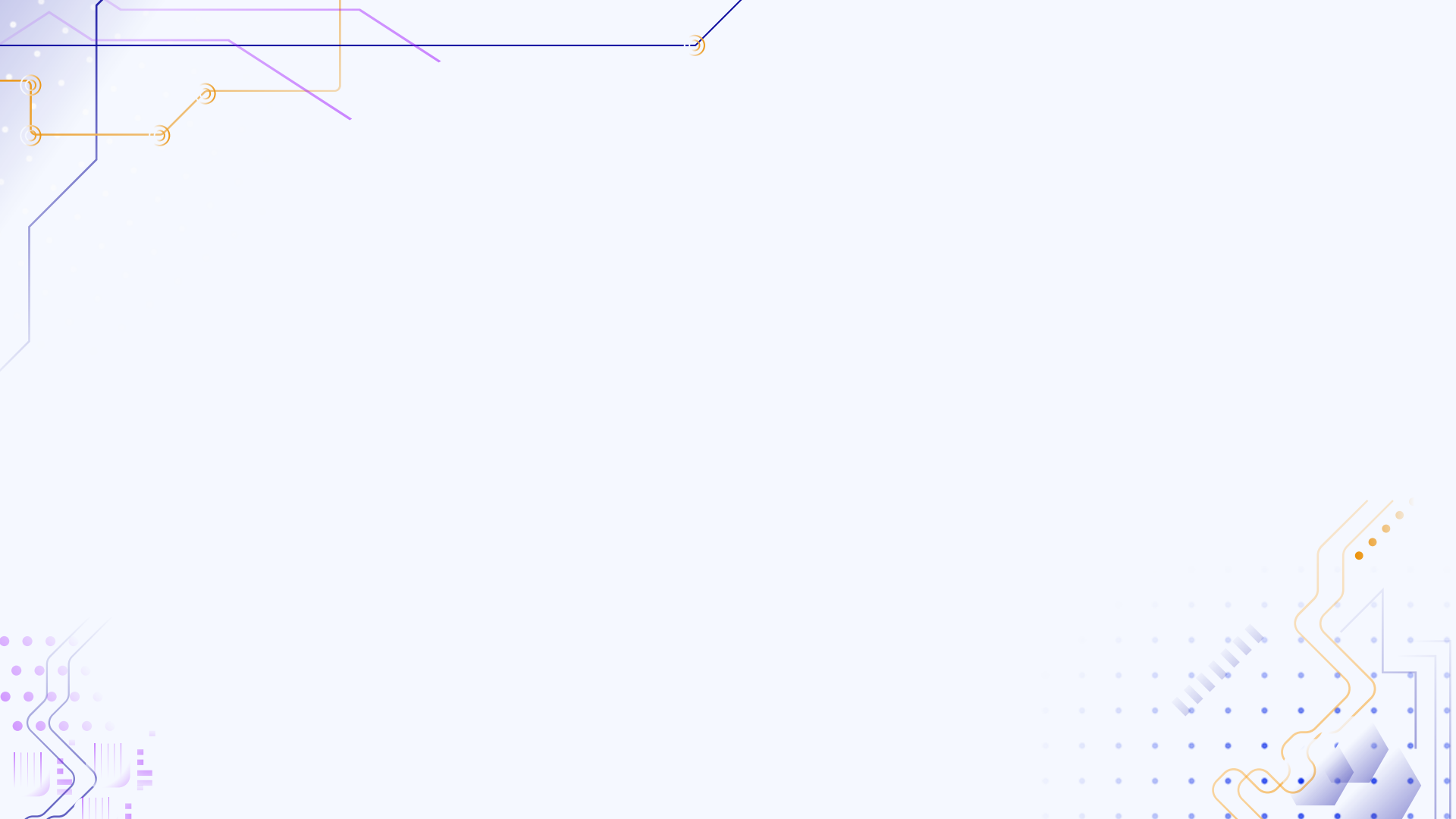


Mettre à disposition les applications pour leur déploiement en production

Dépôt des applications dans un référentiel
(registry, etc...)

Déployer automatiquement le nouveau code en production

Déploiement des applications
dans un environnement de production



The slide features a light blue background with abstract circuit-like patterns in purple and orange. These patterns include lines, dots, and geometric shapes, primarily located in the top-left, bottom-left, and right-hand corners. The main title is centered in the upper half of the slide.

Introduction to Coding Workshop

Here is where your presentation begins

Contents of this template

You can delete this slide when you're done editing the presentation

<u>Fonts</u>	To view this template correctly in PowerPoint, download and install the fonts we used
<u>Used and alternative resources</u>	An assortment of graphic resources that are suitable for use in this presentation
<u>Thanks slide</u>	You must keep it so that proper credits for our design are given
<u>Colors</u>	All the colors used in this presentation
<u>Icons and infographic resources</u>	These can be used in the template, and their size and color can be edited
Editable presentation theme	You can edit the master slides easily. For more info, click <u>here</u>

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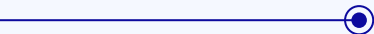


Table of contents

01

Theory lesson

You can describe the topic
of the section here

03

Tips

You can describe the topic
of the section here



02

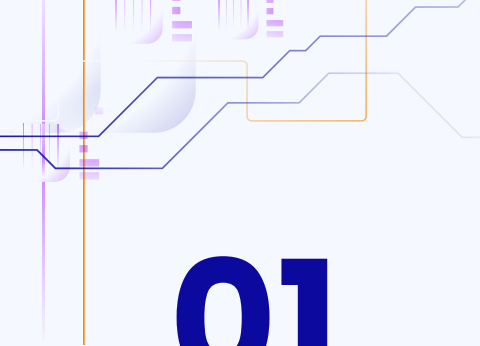
Features of the topic

You can describe the topic
of the section here

04

Practical exercise

You can describe the topic
of the section here



01

Theory lesson

You can enter a subtitle here if you need it



Introduction

Mercury is the closest planet to the Sun and the smallest one in the entire Solar System. **This planet's name has nothing to do with the liquid metal,** since Mercury was named after the Roman messenger god. Mercury's surface is filled with craters

Mercury takes a little more than 58 days to complete its rotation, so try to imagine how long days must be there! **Since the temperatures are so extreme, albeit not as extreme** as on Venus, Mercury has been deemed to be non-habitable for humans



“This is a quote, words full of wisdom
that someone important said and
can make the reader get inspired”

–Someone Famous



Concepts



Mercury

Mercury is the closest planet to the Sun and **the smallest one** in the Solar System—it's only a bit larger than the Moon



Venus

Venus has a beautiful name and is the **second planet from the Sun**. It's hot and has a poisonous atmosphere

What is this topic about?



Mercury

It's the closest planet to the Sun and the **smallest** in the Solar System



Venus

Venus has a beautiful name and is the second planet from the Sun



Mars

Despite being red, Mars is actually a **cold place**. It's full of iron oxide dust

Features of the topic

Mars

Despite **being red**,
Mars is very cold

Jupiter

Jupiter is the biggest
planet of them all

Neptune

It's the farthest
planet from the Sun

Saturn

Saturn is a **gas giant**
and has several rings

Examples



Mercury

It's the closest planet to the Sun and the **smallest** in the Solar System



Venus

Venus has a beautiful name and is the second planet from the Sun



Mars

Despite being red, Mars is actually a **cold place**. It's full of iron oxide dust

Recommendations



Mars

Despite being red,
Mars is very cold



Venus

Venus is the second
planet from the Sun



Neptune

Neptune is the farthest
planet from the Sun



Mercury

Mercury is the closest
planet to the Sun



Saturn

Saturn is a gas giant
and has several rings



Jupiter

Jupiter is the biggest
planet of them all

Image always reinforce the concept

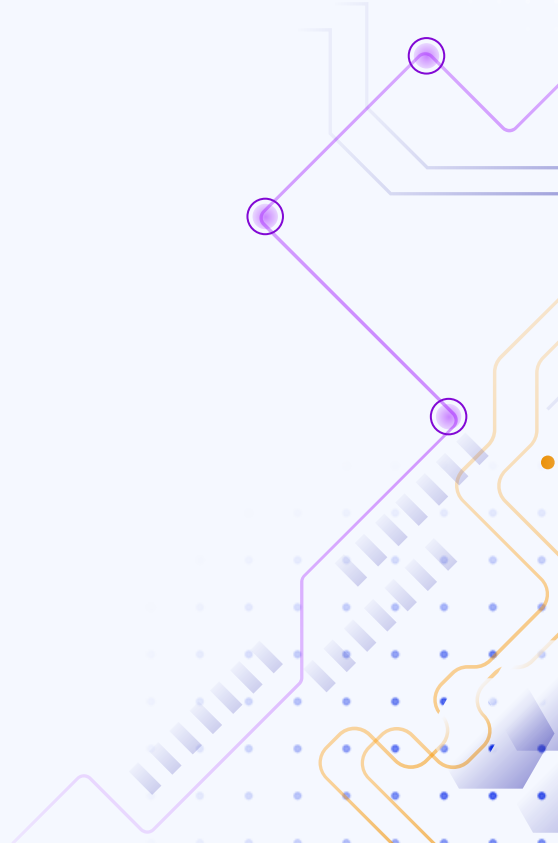
You can give a brief description of the topic you want to talk about here. For example, if you want to talk about Mercury, you can say that it's the smallest planet in the entire Solar System





4,498,300,000

Big numbers catch your audience's attention





9h 55m 23s

Jupiter's rotation period



333,000

The Sun's mass compared to Earth's

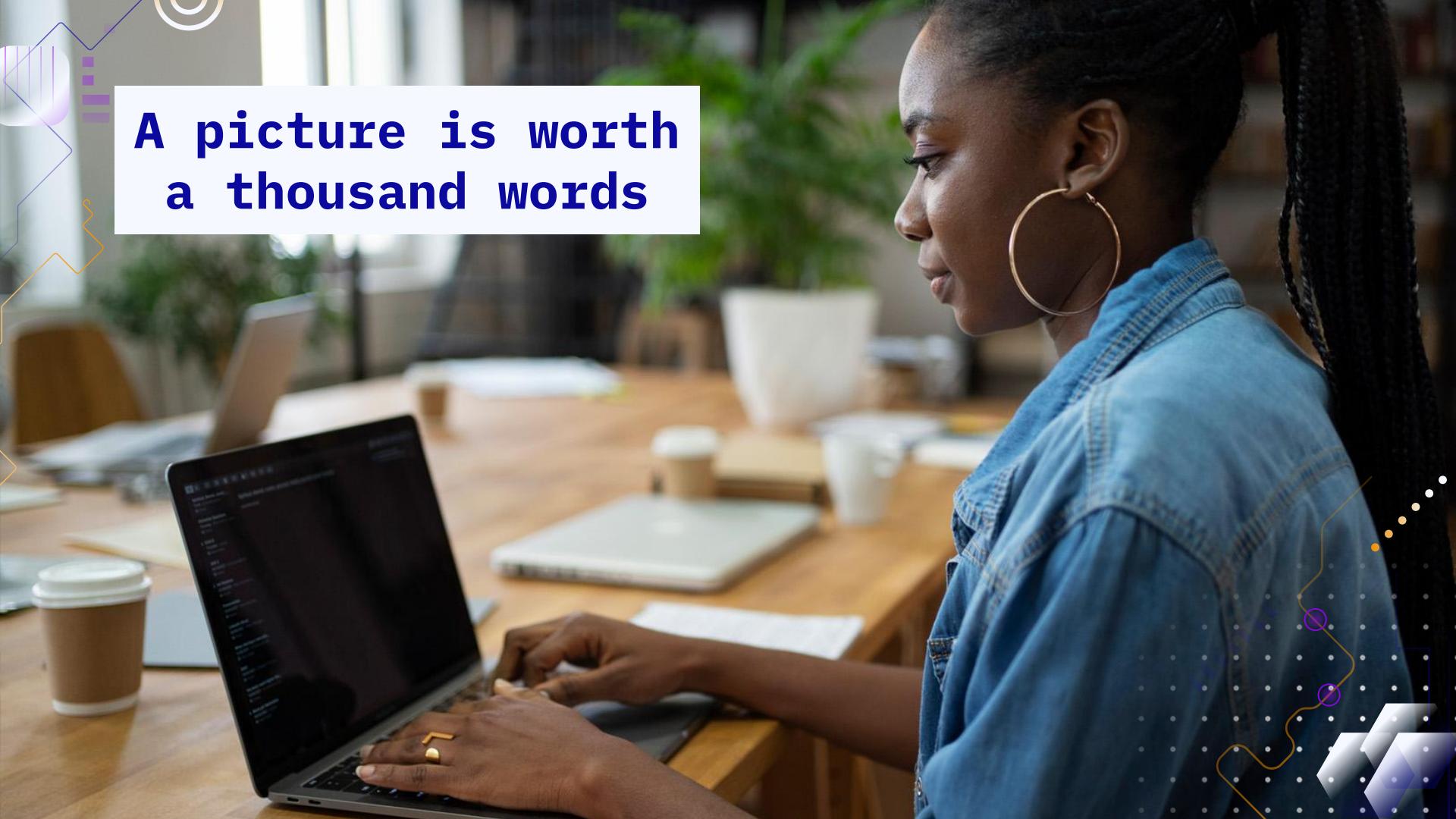
386,000 km

Distance between the Earth and the Moon





Awesome words

A woman with dark skin and long braids is shown in profile, working on a laptop. She is wearing a blue denim shirt and large gold hoop earrings. The setting is a modern office with a wooden desk, a coffee cup, and a potted plant in the background. The image is decorated with geometric line art in the corners.

**A picture is worth
a thousand words**

Practical exercise - calculator

Objective:

Introduce participants to basic coding concepts by building a **simple calculator**

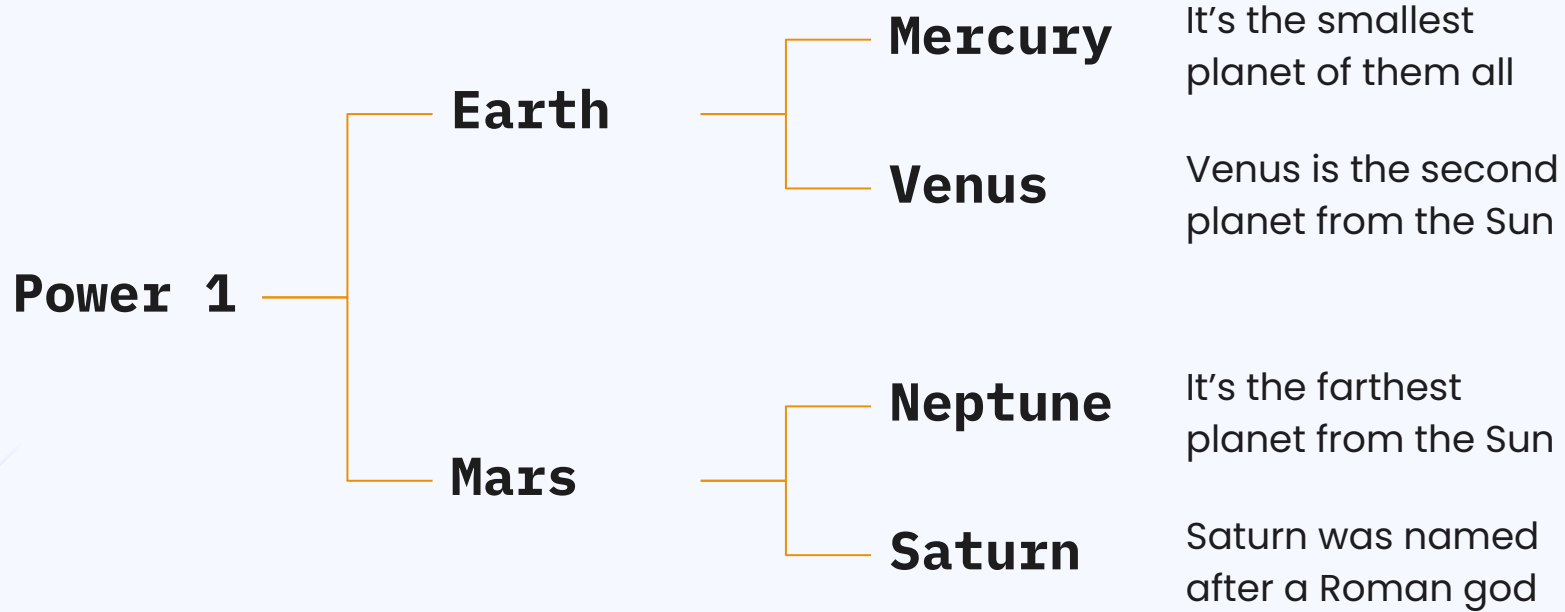
Instructions:

1. Open a Python development environment and write the following code:

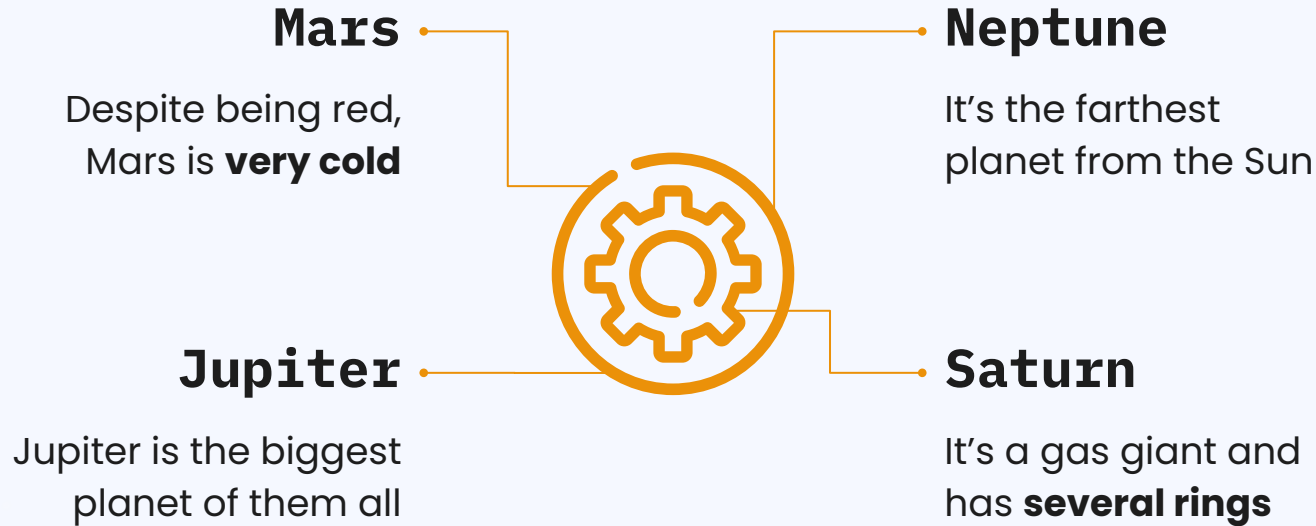
```
# Simple Calculator
num1 = int(input("Enter the first number: "))
num2 = int(input("Enter the second number: "))
print("Sum:", num1 + num2)
print("Difference:", num1 - num2)
print("Product:", num1 * num2)
print("Quotient:", num1 / num2)
```

2. Run the program and experiment with different numbers
3. Observe the output

Brainstorm and idea generation



Main topic and details



Popular programming languages

01 — **Neptune** — Mercury is the closest planet to the Sun and the **smallest** of them all

02 — **Venus** — Venus has a beautiful name and is the **second planet from the Sun**

03 — **Earth** — Earth is the third planet from the Sun and the only one that harbors life in the Solar System

04 — **Saturn** — Saturn is a gas giant and has several rings. It's composed mostly of hydrogen and helium

Sequences

Saturn is composed of
hydrogen and helium

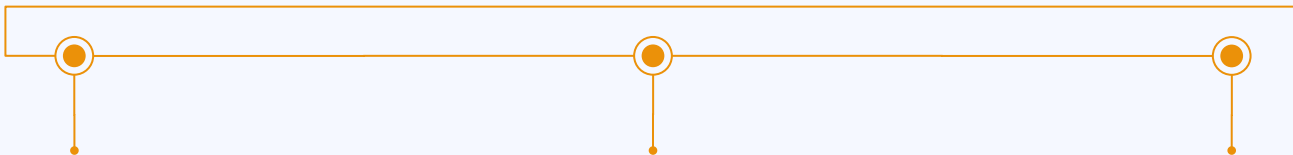
Mercury is the **closest**
planet to the Sun

Jupiter was named
after a Roman god

First

Next

Next



Next

Next

Last

Despite being red,
Mars is **very cold**

Earth is the third
planet from the Sun

Venus has extremely
high temperatures

Classification

Mars	Venus	Mercury	Jupiter
• Small • Red • Cold • Rocky	• Small • Hot • Dry • Volcanic	• Small • Hot • Rocky • Cratered	• Large • Cold • Gassy • Striped
Mars is full of iron oxide dust	Venus has high temperatures	Mercury is quite a small planet	Jupiter is a huge gas giant

Cause and effect

Problem

• Mars

Despite being red,
Mars is very cold

• Venus

Venus is the second
planet from the Sun

Solution

• Mercury

Mercury is the closest
planet to the Sun

• Saturn

Saturn is a gas giant
and has several rings

Question and answer

Question

Is Mercury the closest planet to the Sun and the smallest one in the Solar System? **Note that it's a bit larger than the Moon**

Answer

Venus has a beautiful name and is **the second planet from the Sun**. It's hot and has a poisonous atmosphere

Step-by-step coding

01



Earth

It's the only planet known to **harbor life**

02



Mercury

Mercury is the closest planet to the Sun

03



Jupiter

Jupiter is the **biggest** planet of them all

04



Saturn

Saturn was named after a Roman god

Parts and whole

The whole objective

Mercury is the closest planet to the Sun and the smallest one in the entire Solar System

Parts of the object

- Mercury
- Jupiter
- Venus
- Mars
- Earth
- Saturn
- Mercury

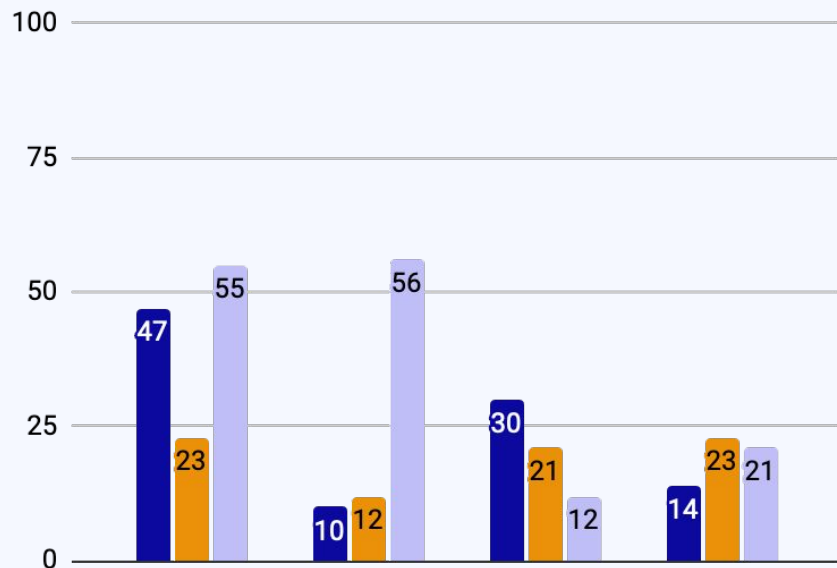
What happens if the parts are missing?

Earth is the third planet from the Sun and the **only one that harbors life in the Solar System**

What's the function of the parts?

Jupiter is a gas giant and the biggest planet in the Solar System

You can use this graph



Follow the link in the graph to modify its data and then paste the new one here. [For more info, click here](#)



Mercury

Mercury is the closest planet to the Sun



Jupiter

Jupiter is the biggest planet of them all



Saturn

Saturn was named after a Roman god

This is a map

USA

Despite being red, Mars is **very cold**

India

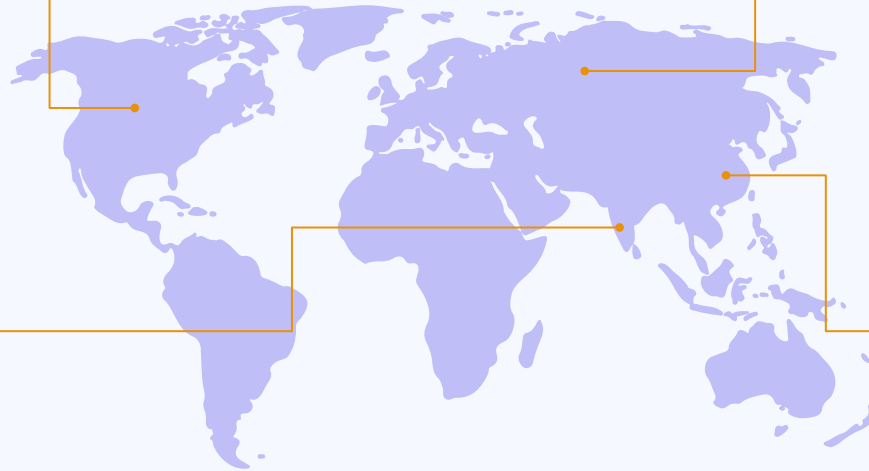
Jupiter is the biggest planet of them all

Russia

Neptune is the farthest planet from the Sun

China

Saturn is **a gas giant** and has several rings



Mockups

You can replace the images on the screen with your own work. Just right-click on them and select “Replace image”



Thanks !

Do you have any questions?

youremail@freepik.com

+34 654 321 432

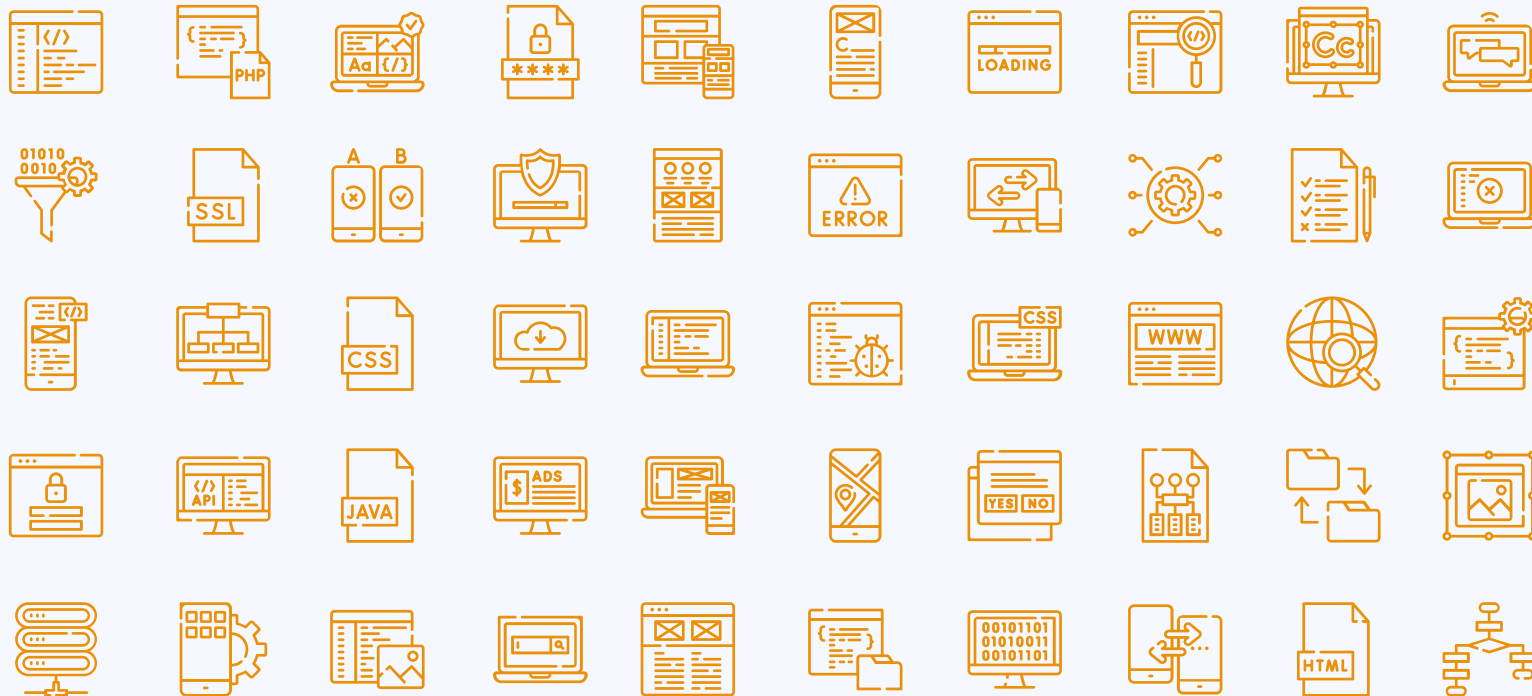
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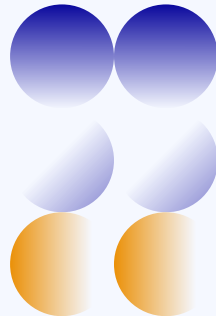
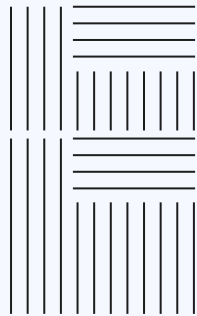
Please keep this slide for attribution

Icon pack



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whose style fits that of this template:



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- [Group of friends planning a trip in a cafe](#)
- [Medium shot man working on laptop](#)
- [Side view of men working on laptops at the office](#)
- [Lifestyle of woman in the office](#)
- [Secretary working on laptop](#)

Vectors:

- [Abstract gradient circuit board background](#)

Icon pack:

- [Icon Pack: Coding | Lineal](#)

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#0c0a9e

#8208d5

#1d1d1d

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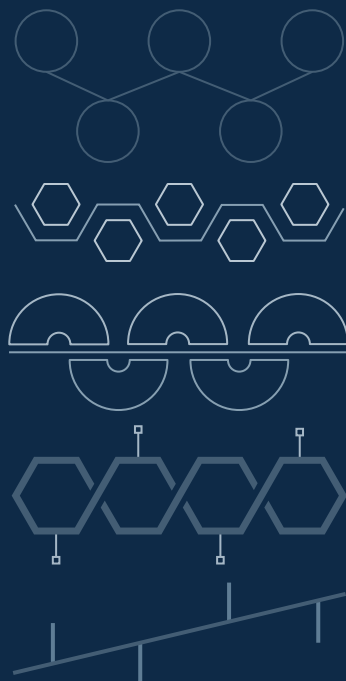
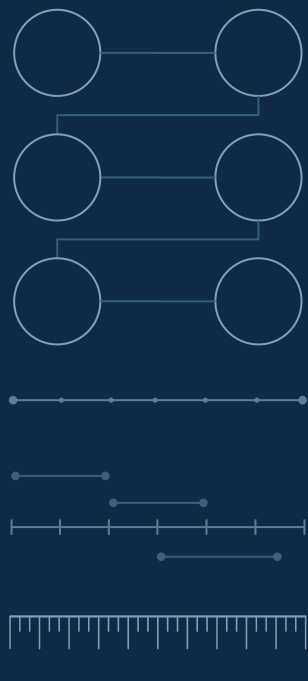
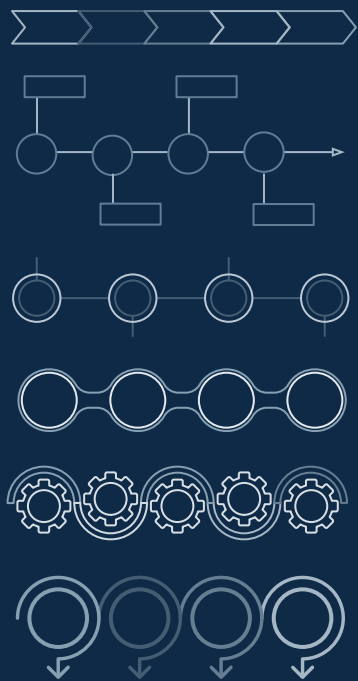
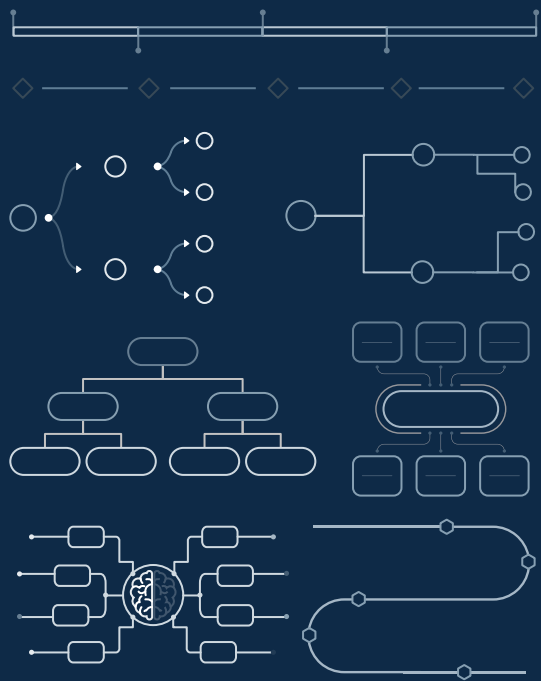
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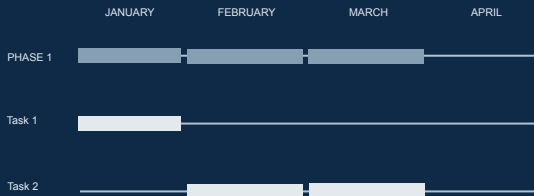
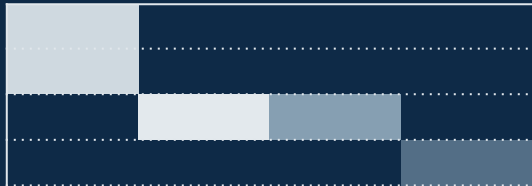
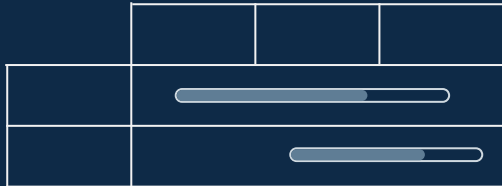
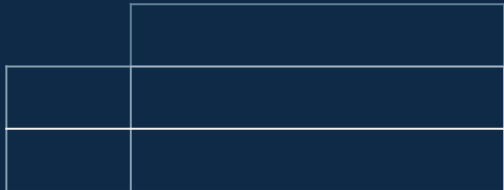
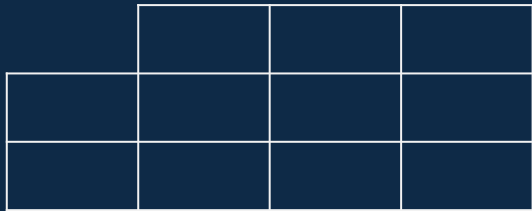
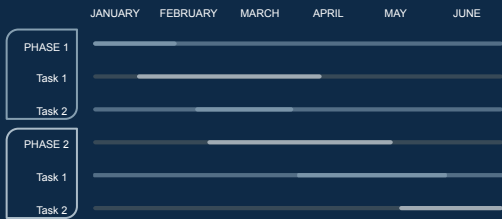
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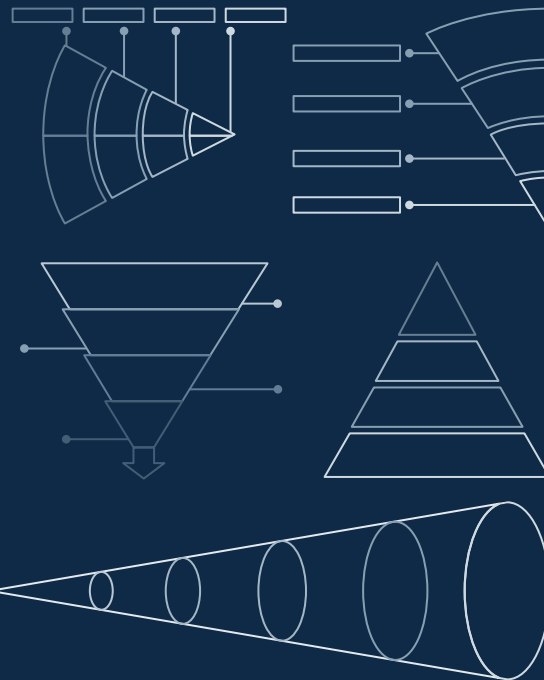
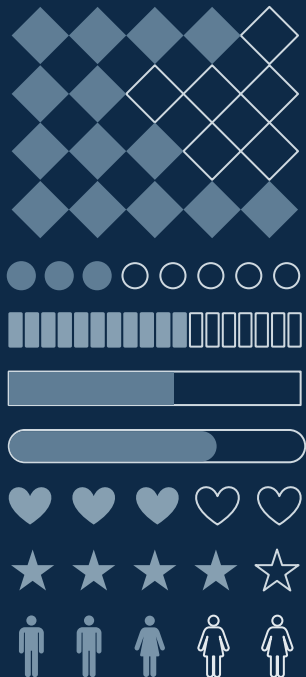
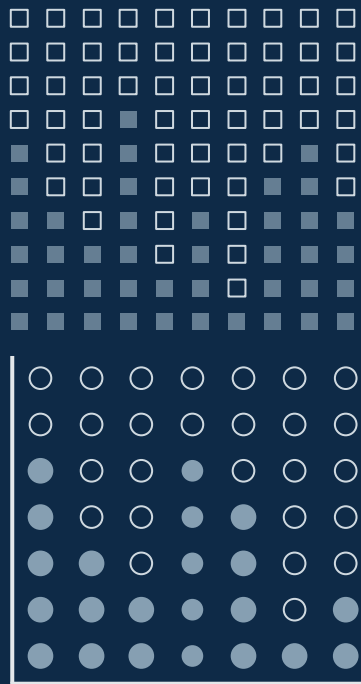












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